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A BRIEF REVIEW ON HAND WRITTEN CHARACTER RECOGNITION

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ABSTRACT

This review describes the state of the art of Character recognition during a period of renewed activity in the field. It is based on an extensive review of the literature, including conference proceedings, journal articles and patents. In the field of pattern recognition, character recognition has gained lot of attention due to its application in various fields. Many researchers have proposed different methodologies for character recognition in different languages. We have reviewed several techniques of character recognition in this paper.

Keywords: Character Recognition, HCR, Off-line Handwriting Recognition, Online Handwriting Recognition, Feature Extraction, Classification, Training and Recognition.

INTRODUCTION

Since hand writing styles of people vary to a great extent, it becomes practically great difficult for the computer to recognize the handwritten characters. To achieve this task, handwritten character recognition comes into picture. The various applications of character recognition are in recognizing the characters in bank cheques and transaction forms, 3D object recognition, car plates, automatic text entry for desktop publication, business card reading, library cataloguing, ledgering, automatic sorting of postal mail, ZIP code, automatic invoice processing.

[Dr.P.S.Deshpande, Mrs.Latesh Malik, Mrs.Sandhya Arora], they concluded that the automation of entire Character recognition process requires a high recognition rate, as well as maximum reliability.

[Pranob K Charles and Om Prakash Sharma et.all], they discussed that Character recognition can be classified into two categories: Online character recognition and Offline character recognition. There are two types of character recognition: Handwritten Character Recognition (HCR) and Optical Character Recognition (OCR) based on the type of input image for the recognition system. Optical Character Recognition is an area within the pattern recognition deals with automatic recognition of various characters in a document image. The task of recognition can be broadly separated into two categories: handwritten data and the machine printed data. Machine printed characters are uniform and unique. While handwritten characters are non-uniform and their size, shape depends on the pen used by the writer. Handwriting of same writer also may vary depending on the situation in which person is writing. It is challenging task to design system, which is capable of identifying the characters with graet accuracy. Various writing styles lead to the distortion in patterns from the standard patterns used to train the system, giving false results. A strong generalized method is required to identify these distorted patterns. As per [Ashlin Deepa R.N, et.all, Supriya Deshmukh,et.all] character recognition systems consist of four major phases: Pre-processing, Segmentation, Feature Extraction, and Classification. They [Om Prakash Sharma,et.all J.Pradeep, et.all, Anshul Gupta, et.all] discussed many classification techniques of character recognition such as template matching , artificial neural networks (ANN) discussed by [Pranob K Charles, et.all, J.Pradeep, et.all, Gaganjot Kaur, et.all, Stuti Asthana, et.all, Sandhya Arora, et.all,],syntactical analysis, wavelet theory, hidden Markov models (HMM) discussed by [Ismael Ahmad Jannoud, M. Antony Robert Raj, et.all, Muhammad Naeem Ayyaz, et.all], [Rajbala Tokas, Aruna Bhadu, Hsin-Chia Fu, Yeong Yuh Xu,] discussed Bayesian theory and [Dileep Kumar Patel, Tanmoy Som, Sushil Kumar Yadav] disussed minimum distance classifiers etc. A lot of methods have been developed for character recognition, among them the neural network is mainly used due its classification efficiency and ease of use [Rajbala Tokas, Aruna Bhadu]. Research has been made in the field of HCR considering varieties of languages like English, Devnagari, Arabic, Thai, Chinese, Tamil and many more. [J.Schuermann, L.D. Harmon,G. Dimauro] discussed that Multiscale Training Technique (MST) is used in many places to solve the generalization problem and its Results are depends largely on resolution of the character images. Image resolution and the training speed have to be optimized to achieve the more percentage of accuracy. Work has been done on character identification in Devnagri script by combining multiple feature extraction techniques like shadow feature, intersection, chain code histogram and straight line fitting. [C.E. Dunn and P.S.P. Wang], discussed another methodology towards feature extraction technique to calculate only twelve directional feature inputs depending upon the gradients, where the hand written character features are the directions of the pixels with respect to their neighboring pixels. [E. Lecolinet and O. Baret], they applied Hybrid method i.e combination of prototype learning/matching method and support vector machines to recognize the hand written characters. [G. Lorette and Y. Lecourtier], they used K-nearest neighbor methods to recognize the patterns. [C.C. Tappert,et.all, D.G. Elliman,et.all, H. Fujisawa,et.all],they discussed that Artificial Neural Nets (perceptron learning) can be used to train the nets and later using the nets identifying the characters, but obtaining 100% accuracy is still a challenge for many such nets. [Rohini B. Kharate, Dr.S.M.Jagade, Sushilkumar N. Holambe], they implemented called Row-wise segmentation technique. This technique (RST) helps in minimization of errors in pattern recognition due to different handwriting styles to great extent.

The structure of the paper is as follows: The working principle of handwritten character recognition followed by literature survey of the same. Conclusions are made in the later part. In this paper a concept of recognizing handwritten character pattern has been developed.

II. THE WORKING PRINCIPLE

In this section, we focus on the methodologies of CR systems. These people [Mr.Narendrasing.B Rajput, Prof.S.M Rajput, Prof.S.M.Badavea] discussed that a hierarchical approach for most of the systems would be from pixel to text, as follows: Pixels ➡ Features ➡ Characters ➡ Subwords ➡ Words ➡ Meaningful Text. And also they said that this bottom up approach varies a great deal, depending upon the type of the CR system and the methodology used. The literature review in the field of CR indicates that the hierarchical tasks are grouped in the stages of the CR for pre-processing, segmentation, representation, training and recognition, post processing. Fig 1 shows block diagram of general character recognition system.

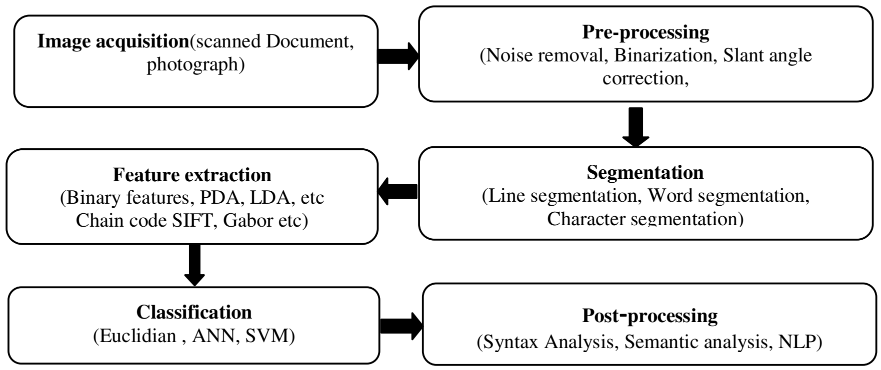

Fig 1: Block Diagram of general Character Recognition System

Image acquisition: The input image for the recognition system can be acquired through a scanner in the form of a scanned image [Anita Rani, Rajneesh Rani, Renu Dhir]. Hence, input to the system is the handwritten characters format and after image acquisition a digital form of the input characters is achieved in the form of image as output.

Pre-processing

Preprocessing aims to produce data that are easy for the character recognition systems to function accurately. It includes removal of unwanted noise and inconsistent data. As these authors [M. Antony Robert Raj,et.all, Rajib Lochan Das, et.all] discussed the sub-process involved in preprocessing are binarization, noise reduction, normalization and skew correction, thinning and slant removal. They discussed that [Om Prakash Sharma, et.all] binarization refers to the conversion of the scanned image into binary image and it is also known as thresholding or digitization. Noise may be added due to poor scanning. As they said [Stuti Asthana,et.all, Paulpandian T and Ganapathy V, Sigappi A.N, et.all] various filters such as Median filtering, Wiener filtering, Butterworth low pass filter, Gaussian low pass filtering and morphological operations may be used for noise removal. As per discussion made in [M. Antony Robert Raj, Dr.S.Abirami] we can convert a random sized image into an standard size image using normalization process. As per [Om Prakash Sharma, et.all] their discussion thinning process may be used for correctness of image and for preservation of endpoints. To estimate deviation of the strokes from the vertical direction, depending on the writing styles, correction of the slant angle is necessary.

Segmentation

The separation of clear character area from non-character area in the given image is the Segmentation. It includes thresholding of the image, skeletonization of the image and pruning [Gaganjot Kaur, Monika Aggarwal]. Skeletonization is the process to reduce the character area to just one pixel line. They discussed [Gaganjot Kaur, Monika Aggarwal] during skeletonization process some unwanted branches called spurs may develop on the output skeletons and these spurs are removed with the help of pruning.

Feature Extraction: To extract the unique features of individual characters, feature extraction techniques like Principle Component Analysis (PCA), Linear Discriminant Analysis (LDA), Independent Component Analysis (ICA), Chain Code (CC), Scale Invariant Feature Extraction

(SIFT), zoning, Gradient based features, Histogram might be applied. These features are used to train the system. As per their discussion [Stuti Asthana, et.all] in feature extraction stage every character is assigned a feature vector to identify it is used to distinguish the character from other characters. According to their discussions [M. Antony Robert Raj, et.all, Muhammad Naeem Ayyaz,et.all, Rajbala Tokas, et.all, Dr. P. S. Deshpande, et.all] feature Extraction techniques can be categorized into three categories. 1. Statistical methods 2. Structural methods 3. Global Transformation and Series Expansion Features.

Classification: Main decision making stage of character recognition system is Classification. It uses the features extracted in the previous stage to identify the text segment according to predefined rules. They [Anshul Gupta,et.all, Sandhya Arora,et.all, Muhammad Naeem Ayyaz,et.all, R. Jagadeesh Kannan,et.all, Anita Rani,et.all] worked on Various classifiers such as Artificial immune system, Associative Memory, Kohonen Network, Support Vector Machine, whereas Nearest Neighbour and K-nearest neighbours are discussed by [Christopher Kermorvant, et.all, M. Antony Robert Raj, et.all, Rakesh Kumar Mandal, et.all] and Bayesian classification, Linear Discriminant Function, Projection Distance, Subspace method, Modified quadratic discriminant function, Mirror Image Learning, Euclidean Distance, Modified Projection distance, Compound projection distance, Compound modified quadratic discriminant function, Regularized Discriminant Analysis are discussed by [Rajbala Tokas,et.all].

Post processing: The classification stage output may contain some errors and these errors can be removed or reduced by post processing methods. The recent research review of the character recognition indicates minor improvements, when only shape recognition of the character is considered. Therefore, they [Mr.Narendrasing.B Rajput, Prof.S.M Rajput, Prof.S.M.Badave] conluded that incorporation of context and shape information in all the stages of character recognition systems is necessary for meaningful improvements in recognition rates.

III. LITERATURE REVIEW

Lot of work has been done in the field of character recognition considering many different languages using various methods and algorithms. [Rajbala Tokas, Aruna Bhadu] have discussed various types of classification of feature extraction methods like statistical feature based methods, structural feature based methods and global transformation techniques. Extracted feature can be either low level or high level. Low level features include character width, height, curliness, aspect ratio etc. As these [Amritha Sampath,et.all] authors discussed that these alone cannot be used to distinguish one character from another in the character set of the language. Hence, there are number of other high level features which include number and position of loops, straight lines, headlines, curves etc. These authors [Gaganjot Kaur, Monika Aggarwal] have selected seven features, like number of boundaries, extent, filled area, length of minor axis and major axis, orientation and solidity and have used Levenberg-Marquardt back propagation training algorithm to train the neural network. They could achieved maximum accuracy of 93% for 130 samples by using 50 hidden layer neurons. He [Tirthraj Dash] has discussed hand character recognition using associative memory net (AMN) and directly worked at pixel level. He was designed Dataset in MS Paint 6.1 with normal Arial font of size 28 and Dimension of image was kept 31 X 39. After characters are extracted, their binary pixel values are directly used to train AMN. These authors [Om Prakash Sharma, M. K. Ghose, Krishna Bikram Shah] have considered Euler number computation to categorize the characters and further proposed an improved zone based hybrid feature extraction model that could achieve higher accuracy with reduced time for training and classification as compared to the diagonal based feature extraction model which gave 98.5% accuracy. They [Sumit Saha, Tanmoy Som] formed weight matrix using the learning rule for neural network and calculated recognition score. Their system had high accuracy rate but mismatch occurred for similar type of characters. Therefore, Euclidean distance measure was further used to recognize such class of characters.

They [Imaran Khan Pathan et al] have proposed offline approach for handwritten isolated Urdu characters and they have used moment invariants (MI) feature to recognize the characters. These MI features are invariant under rotation, translation, scaling and reflection and also these are measure of the pixel distribution around the center of gravity of character and it captures the global character shape information. Their proposed system claim to get highest accuracy of 93.59 %.

They [Rajbala Tokas et al.] have experimented on various feature extraction methods and they concluded that among those methods, cross-corner, diagonal and direction methods are the more accurate and whereas other methods can be combined to obtain higher accuracy rate. These [Rakesh Kumar Mandal et al.] authors have used Row-wise segmentation technique for feature extraction and Perceptron learning method for character recognition and an overall accuracy of 80% was obtained. They have [Dileep Kumar Patel,et.all] used wavelet transform multiresolution technique for feature extraction and Euclidean distance metric was used for character recognition and an average accuracy of 90% was achieved.

These authors [J. Pradeep et al.] have used horizontal based, vertical based and diagonal based feature extraction for character recognition and used feed forward back propagation neural network for classification. Among three feature extractions, the diagonal based feature extraction has been found to be the most accurate. A system to recognize handwritten characters has been developed by [Dr. P. S. Deshpande, et.all] by using histogram band information of the character string and multilayer perception neural network and experimental results shows that some patterns correctly classified with 100 % accuracy whereas remaining number of patterns were classified with absolute relative error.

They [Ashlin Deepa R.N,et.all] have recognized handwritten characters through multiset of puzzle pieces and four basic features like horizontal stroke, vertical stroke, left slant stroke, right slant stroke for labelling purpose and for classification - Exact matching and Probability matching and they have concluded that probability matching is better than exact matching. They [Anita Pal et al.] developed a system to recognize handwritten characters by using the boundary tracing technique for feature extraction and the experimental results show that 94% of recognition accuracy using Fourier descriptors with back propagation network.

[Muhammad Naeem Ayyaz et al.] have used combination of correlation based features and some statistical/structural based features like end points and junction points, invariant moments, projection histogram, profiles and multiclass SVM classification and obtained 96.5% accuracy for digits and 96% for alphabets. In [Rajib Lochan Das,et.all], they used multiple level hidden markov model for classification to recognize the characters by finding gradient features, projection features, curvature features and 98.26% of an average accuracy has been obtained whereas for a significant number of letters, the accuracy rate was close to 100%.

They [Anshul Gupta et al.] have discussed character recognition technique which uses SVM classifiers with four Fourier features and achieved 98.75% accuracy. An efficient fuzzy method was used by [Romesh Ranawana,et.all] for handwritten character recognition which shows 95% of recognition rate if the character is written by the same person who presented the characters to the system during training and recognition rate of around 70% for any other person. They [Velappa Ganapathy et al.] have experimented neural network training on different characters with and without using selective thresholding minimum distance technique and concluded that with selective thresholding minimum distance technique much higher percentage accuracies may be obtained. They proposed [S. Arora et al.] multiple classifier system in which chain code histogram features and moment based features for Devnagari characters extracted. The recognition accuracy of 88.19% was obtained from chain code histogram based classifier and from moment based classifier was 65.67%, whereas by combining both the classifiers using weighted majority scheme gave 98.03% accuracy.

In [Dr. P. S. Deshpande,et.all], they discussed about identification of Devnagari characters by regular expression matching, if they did not match with any pattern they were passed to minimum edit distance filter and overall accuracy of 82% was achieved. [Stuti Asthana et al.] have developed a system using combination of five different scripts and two approaches for recognition in feed forward neural network – one with single hidden layer and other with two hidden layers and achieved 96.53% average recognition rate using double hidden layer and 94% using single hidden layer. [Kai Ding, et.all], they experimented and made a comparative study of Gabor feature against Gradient feature for handwritten Chinese character recognition and it was observed that Gradient feature performed much superior to Gabor feature.

[Ismael Ahmad Jannoud], he has proposed Arabic Handwritten Text Recognition system using Discrete Wavelet Transform technique for feature extraction and they obtained the best recognition rate of 99% for the isolated characters, while the middle characters the worst recognition rate of 91%. In [Ravi Sheth, et.all], these authors have proposed handwritten character recognition system using correlation coefficient and they concluded that it gives good result at the cost of extensive computation of correlation. [P. Phokharatkul et al.], they have developed a system for the recognition of Thai characters and 97% of the training set had been recognized using Ant-Miner Algorithm.

They, [Alex Graves et al.], have introduced a novel approach for both offline and online HCR using Recurrent Neural Network and compared the result using HMM based system. It has been concluded from the experiments that the new approach outperformed with the HMM based system and was more robust to the changes in sample size. They, [Ravi Sheth, et.all], proposed a system using Chain code based approach and the accuracy of chain code feature is tested against feed forward back propagation neural network and support vector machine. It is found that SVM outperforms ANN.

IV. CONCLUSION

In this study, we have overviewed the basic approaches in the character recognition domain attempting to bring out the present status of research in character recognition. From the literature survey that we have made several conclusions have been drawn regarding the feature extraction and the classification techniques. We have found that hybrid feature extraction technique (combining correlation based features and some statistical/structural features) produces results with more accuracy than a single technique. If the number of hidden layers is increased in neural network classification, the recognition rate accuracy increases but there will be an increase in number of epochs and slow processing speed. Whereas SVM classification gives high efficiency with respect to speed, memory and classification accuracy. Row-wise segmentation feature extraction technique is robust yielding better accuracy with fast convergence and comparatively less number of epochs for classification using neural network, if the characters are written in box sheets, not deviating from the actual position. The success rate can be achieved by each method in spite of their own advantages and limitations. However, when we consider various databases, sample spaces and constraints it is difficult to comment about the success of recognition methods, especially in terms of recognition rates. In case of handwritten texts under poor conditions or for free style handwriting, an improved approach in almost all stages of CR research is needed.

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